Nonlinear dynamics: Newmark scheme and Newton–Raphson

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1 Equation of motion

For one degree of freedom, with internal forces depending on the solution, the semi-discrete equation of motion at time step $n+1$ reads

$$M\ddot{u}^{n+1} + C\dot{u}^{n+1} + f^{\text{int}}(u^{n+1}, \boldsymbol{\alpha}^n) - f^{\text{ext}}(u^{n+1}, t^{n+1}) = r(u^{n+1}) = 0, \tag{1}$$

where $\boldsymbol{\alpha}^n$ denotes the internal (history) variables, known at the end of step n and frozen during the solution of step $n+1$ (for a damage model: the damage d reached so far).

2 Newmark β scheme in predictor–corrector form

For a time step $\Delta t = t^{n+1} - t^n$ (Belytschko), the displacement and velocity at $n+1$ are

$$u^{n+1} = \tilde{u}^{n+1} + \beta\Delta t^2\,\ddot{u}^{n+1}, \qquad \tilde{u}^{n+1} = u^n + \Delta t\,\dot{u}^n + \frac{\Delta t^2}{2}(1-2\beta)\,\ddot{u}^n, \tag{2}$$

$$\dot{u}^{n+1} = \tilde{\dot{u}}^{n+1} + \gamma\Delta t\,\ddot{u}^{n+1}, \qquad \tilde{\dot{u}}^{n+1} = \dot{u}^n + (1-\gamma)\Delta t\,\ddot{u}^n. \tag{3}$$

Inverting (2) and substituting in (3):

$$\ddot{u}^{n+1} = \frac{1}{\beta\Delta t^2}(u^{n+1} - \tilde{u}^{n+1}), \qquad \dot{u}^{n+1} = \tilde{\dot{u}}^{n+1} + \frac{\gamma}{\beta\Delta t}(u^{n+1} - \tilde{u}^{n+1}). \tag{4}$$

Both (4) are substituted into (1), which becomes a nonlinear equation in the single unknown u^{n+1}:

$$r(u^{n+1}) = \Big(\frac{M}{\beta\Delta t^2} + \frac{\gamma\,C}{\beta\Delta t}\Big)(u^{n+1} - \tilde{u}^{n+1}) + C\,\tilde{\dot{u}}^{n+1} + f^{\text{int}}(u^{n+1}) - f^{\text{ext}}(u^{n+1}, t^{n+1}) = 0. \tag{5}$$

With $C = 0$ the damping terms vanish and only the inertia term $M(u^{n+1} - \tilde{u}^{n+1})/(\beta\Delta t^2)$ is added to the static residual.

Properties. $\beta = \frac{1}{4}$, $\gamma = \frac{1}{2}$: average acceleration, second-order accurate, unconditionally stable, no numerical damping. For $\gamma > \frac{1}{2}$ the scheme is first-order accurate and dissipative (algorithmic damping growing with $\gamma - \frac{1}{2}$ and with $\omega\Delta t$); unconditional stability then requires $\beta \geq (\gamma + \frac{1}{2})^2/4$.

3 Newton–Raphson

The residual (5) is linearised around the iterate u_i^{n+1} by a Taylor expansion:

$$r(u_{i+1}^{n+1}) \approx r(u_i^{n+1}) + \frac{\partial r}{\partial u}\Big|_{u_i^{n+1}} \Delta u_i^{n+1} = 0, \qquad \Delta u_i^{n+1} = u_{i+1}^{n+1} - u_i^{n+1}, \tag{6}$$

with the consistent tangent obtained by differentiating (5):

$$\frac{\partial r}{\partial u} = \frac{M}{\beta \Delta t^2} + \frac{\gamma\, C}{\beta \Delta t} + \frac{\partial f^{\text{int}}}{\partial u} - \frac{\partial f^{\text{ext}}}{\partial u}. \tag{7}$$

The last term vanishes when the external force does not depend on the displacement (dead load, imposed displacement), which is the usual case. Solving (6) for the increment and updating:

$$\Delta u_i^{n+1} = -\Big(\frac{\partial r}{\partial u}\Big|_{u_i^{n+1}}\Big)^{-1} r(u_i^{n+1}), \qquad u_{i+1}^{n+1} = u_i^{n+1} + \Delta u_i^{n+1}. \tag{8}$$

For several degrees of freedom, $\partial r/\partial u$ is the effective tangent matrix $\mathbf{K}_{\text{eff}} = \mathbf{M}/(\beta\Delta t^2) + \gamma\mathbf{C}/(\beta\Delta t) + \mathbf{K}_t$ and (8) is a linear solve.

4 Algorithm for one time step

1. Predictors $\tilde{u}^{n+1}$, $\tilde{\dot{u}}^{n+1}$ from (2)–(3).
2. Initial iterate $u_0^{n+1} = \tilde{u}^{n+1}$; internal variables frozen at $\boldsymbol{\alpha}^n$.
3. Iterate (8) until $|r(u_i^{n+1})| \le \varepsilon$ (absolute tolerance) or $\|\Delta u_i^{n+1}\| \le \varepsilon_u \|u^{n+1}\|$ (relative tolerance); stop with an error after $i_{\max}$ iterations.
4. Correctors: $\ddot{u}^{n+1}$ and $\dot{u}^{n+1}$ from (4).
5. Update of the internal variables $\boldsymbol{\alpha}^{n+1}$ (e.g. $d^{n+1} = \max(d^n, \hat{d}(u^{n+1}))$ for a damage model).

At $t = 0$ the acceleration is obtained from the equation of motion itself: $\ddot{u}^0 = M^{-1}(f^{\text{ext}}(u^0, 0) - C\dot{u}^0 - f^{\text{int}}(u^0))$.

5 Application: two springs in series, spring 1 damageable

Node 1 of mass m_1 between a damageable spring $k_1^d(u_1) = k_1(1-d)$, $d = \max \hat{d}(u_1)$ and an elastic spring k_2 whose other end is displacement controlled, $u_2(t)$. Then

$$M = m_1, \quad C = 0, \quad f^{\text{int}}(u_1) = (k_1^d(u_1) + k_2)u_1, \quad f^{\text{ext}}(t) = k_2\, u_2(t), \tag{9}$$

and (5)–(7) become

$$r(u_1) = \frac{m_1}{\beta\Delta t^2}(u_1 - \tilde{u}) + (k_1^d(u_1) + k_2)u_1 - k_2 u_2^{n+1}, \qquad \frac{\partial r}{\partial u_1} = \frac{m_1}{\beta\Delta t^2} + k_1^d + u_1 \frac{\mathrm{d}k_1^d}{\mathrm{d}u_1} + k_2. \tag{10}$$

The static problem is the limit $m_1 \to 0$: $R(u_1) = (k_1^d + k_2)u_1 - k_2 u_2$, with tangent $k_t = k_1^d + u_1\, \mathrm{d}k_1^d/\mathrm{d}u_1 + k_2$, which becomes negative in the softening range when $k_2 < \max(-\mathrm{d}F_1/\mathrm{d}u_1)$. The inertia term $m_1/(\beta\Delta t^2)$ added to the tangent is what restores a strictly increasing residual and a unique root.